Software Architecture (WS 26/27)
Assignment 2

Prof. Dr.-Ing. Ronny Seiger (ronny.seiger@rwth-aachen.de)
Tobias Arens (tobias.arens@rwth-aachen.de)

22.10.2026

Task 1: Finalize your project ideas

We will start implementing first parts of the software architecture.

Task: Please finalize your project ideas and think about example workflows that might involve multiple services and humans interacting with each other.

Task 2: Create your first software components

Task: Implement **two** standalone software components in the form of web services with an HTTP (REST) API. Both services should be able to exchange data via their HTTP APIs.

You can find the example presented in the exercise session here:
`https://gitlab.git.nrw/rwth-swa/teaching/swa-ws26-sample`

Implementation hints:

- Both services should have a simple layered structure, separating the REST controllers (for ingoing and outgoing calls) from the domain/business logic.
- Both services should provide at least one specific business functionality beyond CRUD operations.
- At least one service should be implemented using the Spring Boot framework. We recommend to use Maven as build system.
- There should be at least one point of synchronous HTTP requests between the services.
- Both service should be testable as standalone services via their HTTP interfaces (e.g., using CURL, PostMan, or Bruno).
- Both services should be running in one Docker container each in the same Docker environment.
- There is no need to add a database to the services yet. We will investigate this aspect later.
- During the entire course, user interfaces for the software components are optional.